



National
Qualifications
2025

X813/75/02

Chemistry
Section 1 — Questions

THURSDAY, 1 MAY

1:00 PM – 3:30 PM

Instructions for the completion of Section 1 are given on *page 02* of your question and answer booklet X813/75/01.

Record your answers on the answer grid on *page 03* of your question and answer booklet.

You may refer to the Chemistry Data Booklet for National 5.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



* X 8 1 3 7 5 0 2 *

SECTION 1 — 25 marks

Attempt ALL questions

1. Oganesson is a newly discovered element with a predicted electron arrangement of 2,8,18,32,32,18,8

In which group of the periodic table should oganesson be placed?

- A 0
- B 1
- C 2
- D 7

2. Which line in the table shows the correct number of protons and electrons for the atom, $^{56}_{26}\text{Fe}$?

	Protons	Electrons
A	56	56
B	56	26
C	26	26
D	26	56

3. Which of the following molecules contains three atoms?

- A Phosphorus trichloride
- B Magnesium chloride
- C Sodium fluoride
- D Carbon dioxide

4. Which bonding and structure is never found in elements?

- A Covalent molecular
- B Ionic lattice
- C Metallic lattice
- D Covalent network

5. Which of the following compounds would conduct electricity at 600 °C?
You may wish to use the data booklet to help you.
- A Silicon dioxide
 - B Lithium bromide
 - C Ammonia
 - D Barium chloride
6. Which of the following oxides, when shaken with water, would give an alkaline solution?
You may wish to use the data booklet to help you.
- A Calcium oxide
 - B Nitrogen dioxide
 - C Sulfur dioxide
 - D Nickel oxide
7. A student made some statements about the effect of adding water to a solution of hydrochloric acid.
Identify the correct statement.
- A The rate of reaction with magnesium will increase.
 - B The concentration of H^+ ions will increase.
 - C The acid will be neutralised.
 - D The pH will increase.
8. Ammonium phosphate, $(\text{NH}_4)_3\text{PO}_4$, has a gram formula mass of 149.
The percentage by mass of nitrogen in ammonium phosphate is equal to
- A $\frac{14}{149} \times 100$
 - B $\frac{42}{149} \times 100$
 - C $\frac{149}{14} \times 100$
 - D $\frac{149}{42} \times 100$

[Turn over

9. What is the charge on the manganese ion in manganese dichromate, MnCr_2O_7 ?
You may wish to use the data booklet to help you.

- A 2+
- B 1+
- C 2-
- D 1-

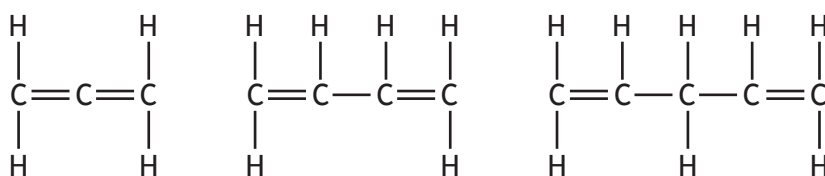
10. Compound X:

- is saturated
- is insoluble in water
- has two hydrogen atoms for every carbon atom.

Which of the following could be compound X?

- A Propane
- B Propan-1-ol
- C Cyclopropane
- D Propene

11. The structures of three members of a homologous series called the dienes are shown.



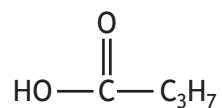
The general formula for the dienes is

- A C_nH_{n+1}
- B C_nH_{n+2}
- C C_nH_{2n}
- D $\text{C}_n\text{H}_{2n-2}$

12. Which of the following formula masses belongs to a hydrocarbon that does **not** belong to the same homologous series as the others?

- A 16
- B 44
- C 58
- D 70

13. The structural formula for a compound is shown.

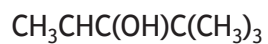


The name of this compound is

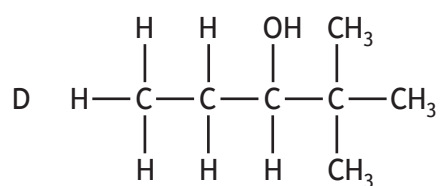
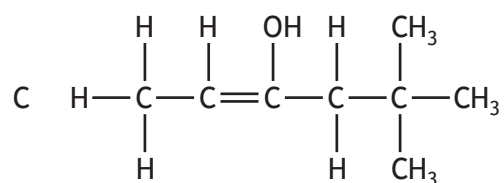
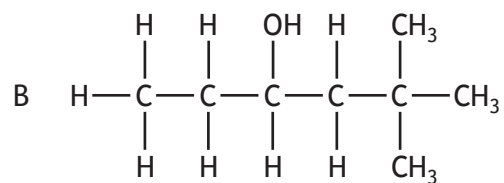
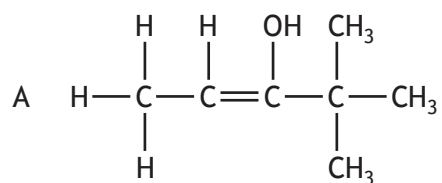
- A propanoic acid
- B propan-1-ol
- C butanoic acid
- D butan-1-ol.

[Turn over

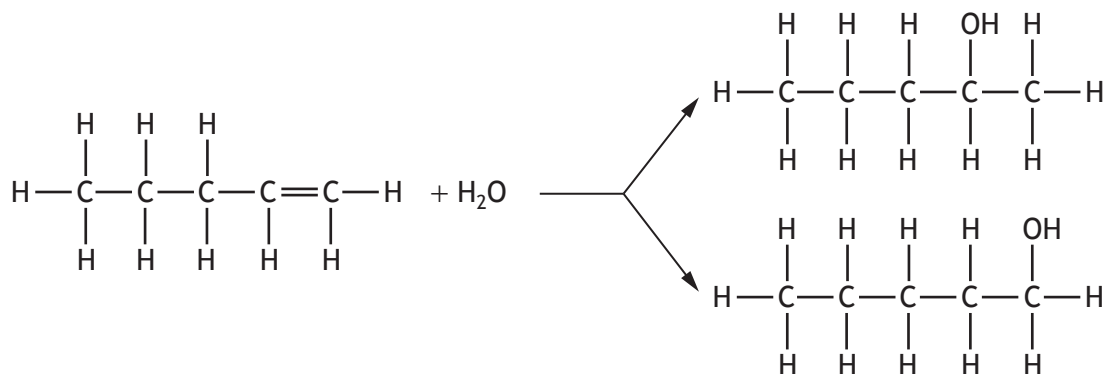
14. The shortened structural formula for a compound is shown.



Which of the following is another way of representing this structure?



15. Pent-1-ene reacts with water to form two products.

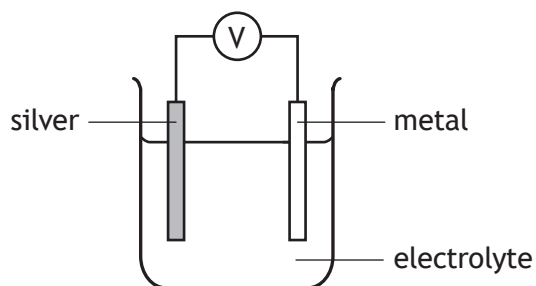


Which of the following alkenes does **not** form two products on reaction with water?

- A But-1-ene
 - B But-2-ene
 - C Hex-1-ene
 - D Hex-2-ene
16. When 5 g of a fuel was burned, 150 kJ of energy was released.
Calculate the energy, in kJ, that would be released if 60 g of the fuel was burned.
- A 30
 - B 150
 - C 1800
 - D 9000
17. Which of the following is **not** involved in metallic bonding?
- A Positive ions
 - B Negative ions
 - C Delocalised electrons
 - D Electrostatic attraction

[Turn over

18. Four different metals were connected to silver in a cell.



The results obtained were recorded in the table.

Metal	Voltage (V)
Iron	0.9
Zinc	1.1
Magnesium	2.7
Metal X	1.5

Which of the following could be metal X?

You may wish to use the data booklet to help you.

- A Aluminium
- B Sodium
- C Nickel
- D Copper

19. Which of the following is **not** a redox reaction?

- A $\text{Zn(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
- B $\text{Cu(s)} + \text{Cl}_2(\text{g}) \rightarrow \text{Cu}^{2+}(\text{Cl}^-)_2(\text{s})$
- C $\text{Br}_2(\text{aq}) + 2\text{Fe}^{2+}(\text{aq}) \rightarrow 2\text{Fe}^{3+}(\text{aq}) + 2\text{Br}^-(\text{aq})$
- D $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{Ba}^{2+}\text{SO}_4^{2-}(\text{s})$

20. Which of the following would be recovered unchanged in the Ostwald process?

- A Water
- B Oxygen
- C Ammonia
- D Platinum

21. In nuclear equations a proton is represented as

- A ${}^1_1\text{p}$
- B ${}^1_0\text{p}$
- C ${}^0_1\text{p}$
- D ${}^0_0\text{p}$

22. Which of the following compounds can be prepared by precipitation?
You may wish to use the data booklet to help you.

- A Lithium sulfate
- B Sodium sulfate
- C Barium sulfate
- D Magnesium sulfate

23. A student was reading the volume of liquid in a piece of apparatus.



The correct volume, in cm^3 , that the student should record is

- A 8.8
- B 9.1
- C 9.2
- D 10.8

[Turn over

Questions 24 and 25 refer to the experiment below.

A titration was performed using hydrochloric acid and a 10 cm³ solution of sodium hydroxide. The results are shown in the table.

Titration	Initial reading (cm ³)	Final reading (cm ³)	Titre (cm ³)
1	0.0	11.0	11.0
2	11.0	21.1	10.1
3	22.0	32.6	10.6
4	33.0	43.3	10.3

24. The average of the concordant titre values, in cm³, is

- A 10.8
- B 10.5
- C 10.3
- D 10.2

25. Which piece of apparatus is the most appropriate to measure and transfer the 10 cm³ of sodium hydroxide solution?

- A Burette
- B Pipette
- C Conical flask
- D Measuring cylinder

**[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF
YOUR QUESTION AND ANSWER BOOKLET]**



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Mark

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X813/75/01**Chemistry
Section 1 — Answer grid
and Section 2**

THURSDAY, 1 MAY

1:00 PM – 3:30 PM



* X 8 1 3 7 5 0 1 *

Fill in these boxes and read what is printed below.

Full name of centre

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Town

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Forename(s)

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Surname

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Number of seat

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Date of birth

Day

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Month

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Year

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Scottish candidate number

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Total marks — 100**SECTION 1 — 25 marks**

Attempt ALL questions.

Instructions for the completion of Section 1 are given on *page 02*.**SECTION 2 — 75 marks**

Attempt ALL questions.

You may refer to the Chemistry Data Booklet for National 5.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



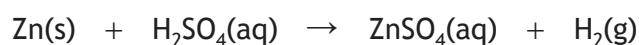
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SECTION 2 — 75 marks

Attempt ALL questions

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1. (a) The reaction of zinc with sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, is shown.



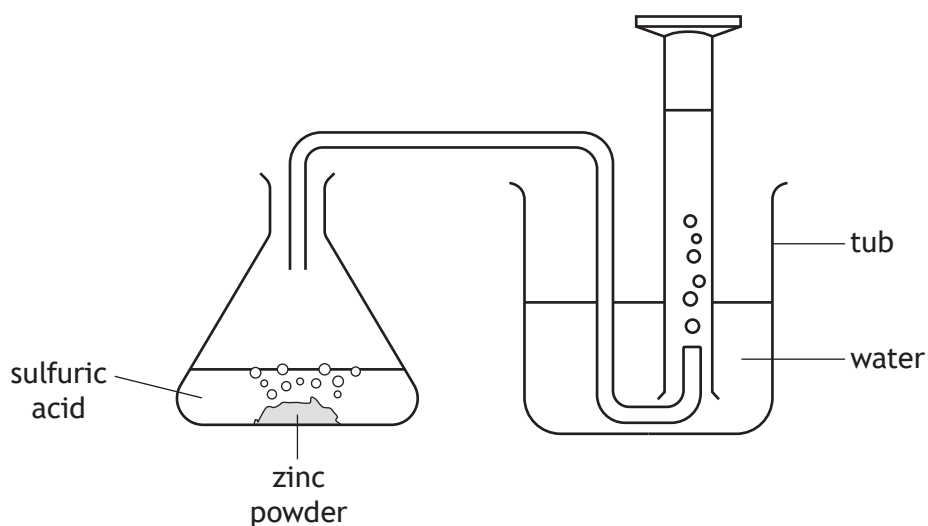
- (i) Hydrogen gas is produced in this reaction.

State the test, including the result, for hydrogen gas.

1

- (ii) A student was asked to draw a diagram of the apparatus used to **collect and measure** the volume of hydrogen gas produced during the reaction.

Their diagram is shown.



Describe **one** way in which the student's diagram is incorrect.

1

- (iii) A solution of zinc(II) sulfate, $\text{ZnSO}_4(\text{aq})$, is produced in this reaction.

Name the technique that could be used to obtain a **dry** sample of zinc(II) sulfate, $\text{ZnSO}_4(\text{s})$.

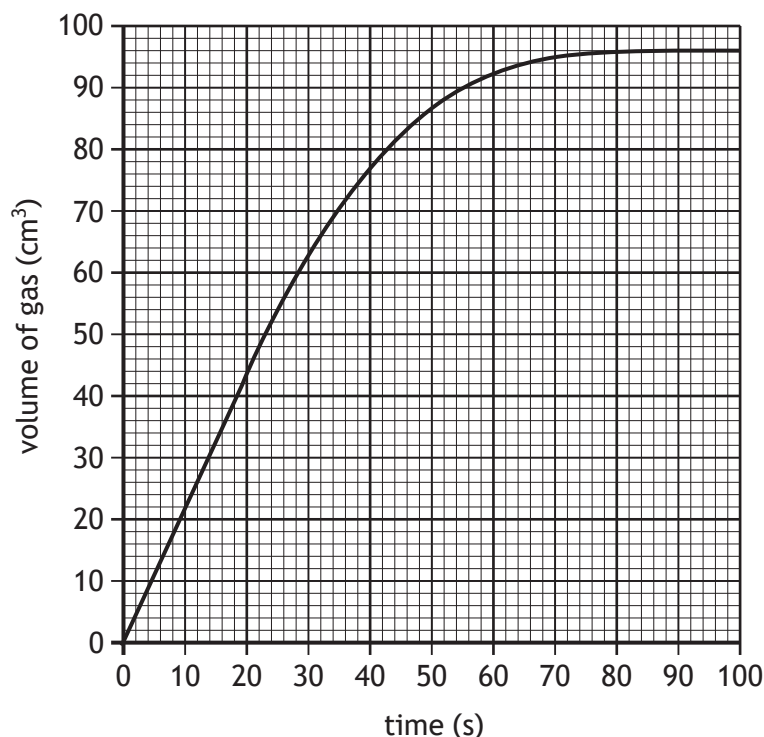
1



1. (continued)

- (b) The volume of hydrogen gas collected over time when zinc powder is added to 100 cm³ of dilute sulfuric acid was recorded.

A graph of the results is shown.



- (i) Calculate the average rate of reaction, in cm³ s⁻¹, for the first 50 seconds. 2
Show your working clearly.

- (ii) **Add a curve to the graph** to show the results that would be expected if the experiment was repeated using the same mass of zinc lumps. 1
(An additional graph, if required, can be found on *page 31*.)

- (iii) A similar experiment was carried out using the same mass of zinc powder and the same volume and concentration of hydrochloric acid, HCl(aq), in place of sulfuric acid, H₂SO₄(aq).
Predict the volume of gas, in cm³, produced in this reaction. 1



2. Oxygen is an odourless, colourless gas that is present in the Earth's atmosphere.

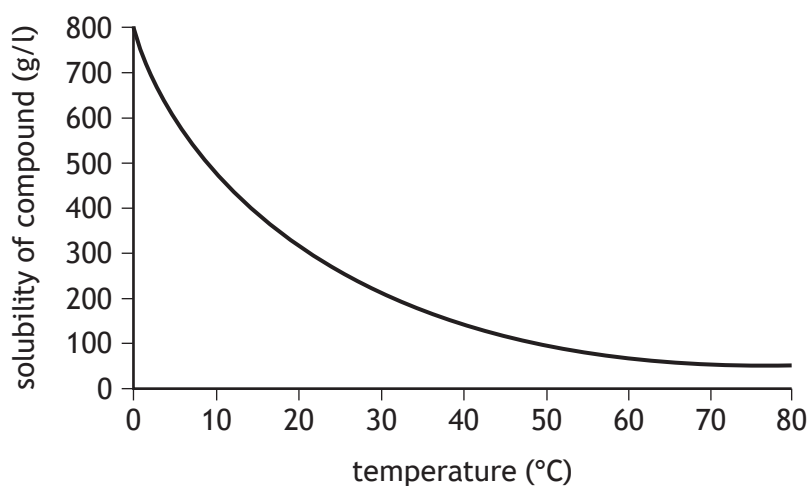
(a) Draw a diagram, showing **all** the outer electrons, to represent a molecule of oxygen, O_2 .

1

(b) Name an element that has similar chemical properties to oxygen.

1

(c) The graph shows the solubility of a compound containing oxygen at different temperatures.



State the relationship between temperature and the solubility of the compound.

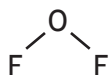
1



2. (continued)

- (d) Oxygen can react with fluorine to form oxygen fluoride molecules, containing covalent bonds.

A molecule of oxygen fluoride is shown.



- (i) State the name used to describe the shape of an oxygen fluoride molecule.

1

- (ii) Circle the correct words to complete the sentence.

1

A covalent bond forms when two $\left\{ \begin{array}{l} \text{positive} \\ \text{negative} \\ \text{neutral} \end{array} \right\}$ nuclei are held together
by their common attraction for a shared pair of $\left\{ \begin{array}{l} \text{protons} \\ \text{neutrons} \\ \text{electrons} \end{array} \right\}$.

[Turn over

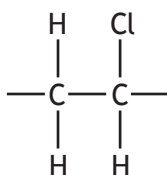


* X 8 1 3 7 5 0 1 0 9 *

3. Plastics which soften when heated are classed as thermosoftening plastics.

- (a) Poly(vinyl chloride), PVC, is an example of a thermosoftening plastic and is used in the manufacture of clothing.

The repeating unit of PVC is shown.



- (i) Draw the monomer used to make PVC.

1

- (ii) When PVC is burned, the gas hydrogen chloride is formed.

Hydrogen chloride molecules, HCl(g) , contain only two atoms.

State the term used to describe a molecule containing only two atoms.

1



* X 8 1 3 7 5 0 1 1 0 *

3. (continued)

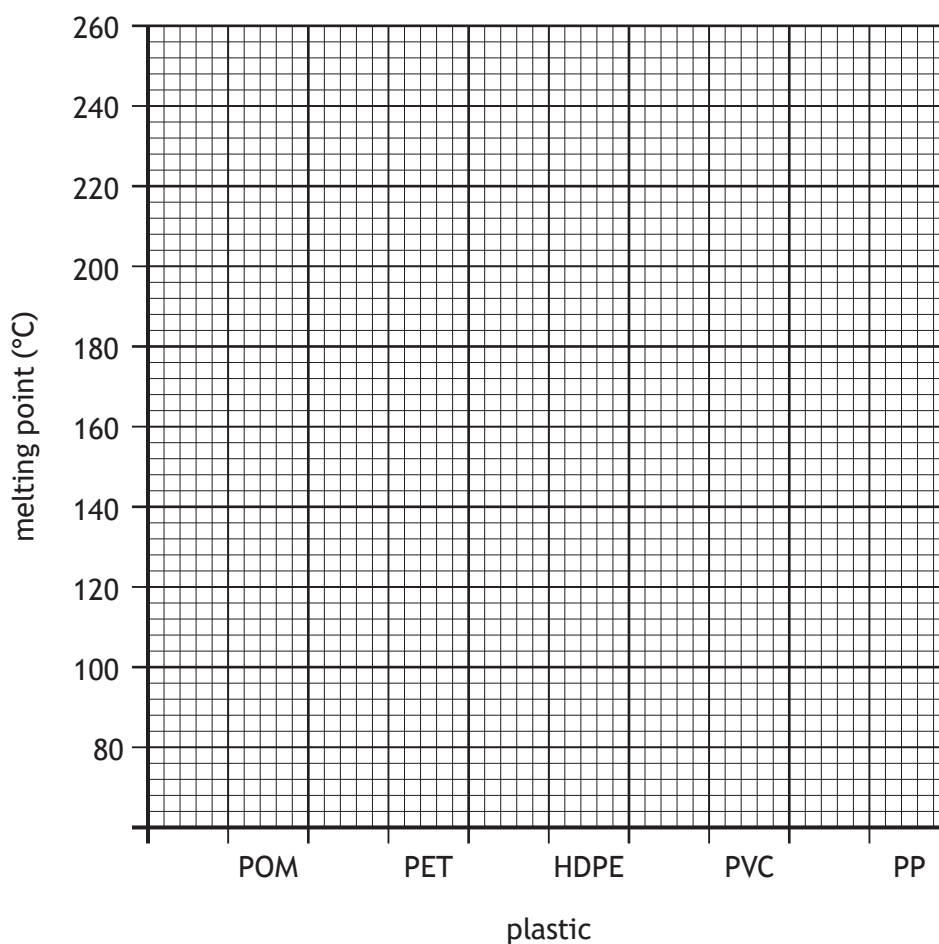
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(b) A table of some thermosoftening plastics and their melting point is shown.

Plastic	Melting point (°C)
POM	165
PET	260
HDPE	131
PVC	82
PP	130

- (i) Complete the graph to show the melting point for each of the plastics.
(An additional graph, if required, can be found on *page 32*.)

2

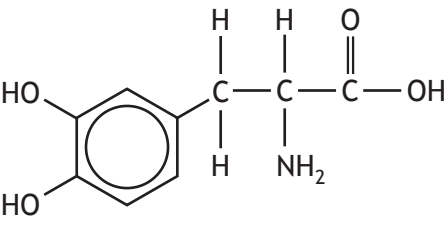
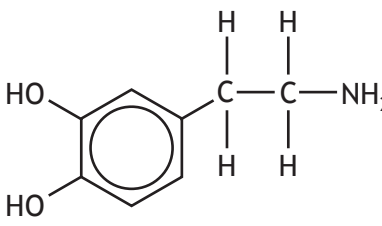
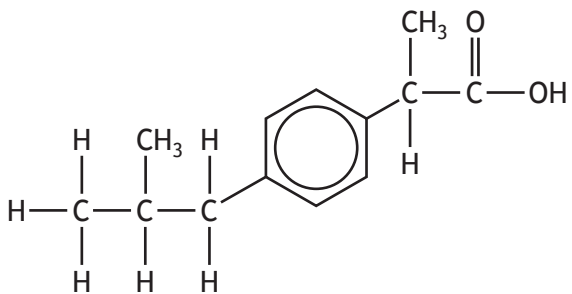


- (ii) Suggest why PVC could **not** be used as a container for boiling water.

1



4. Many medicines are compounds that contain carbon atoms.
The table shows some examples of these medicines and their use.

Name of medicine	Use of medicine	Structure of molecule
L-DOPA	treatment of Parkinson's disease	
Dopamine	treatment of low blood pressure	
Ibuprofen	treatment of pain and inflammation	

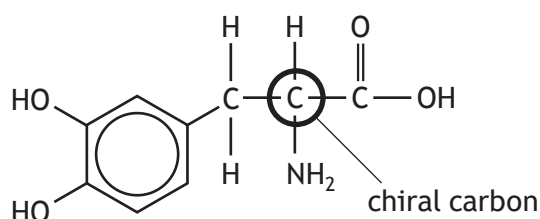
- (a) Name the functional group that is present in both L-DOPA and ibuprofen but is **not** present in dopamine. 1
- (b) A key step in producing L-DOPA involves an addition reaction with hydrogen molecules. This is carried out using a catalyst.
- (i) Name the type of addition reaction that involves the addition of hydrogen. 1
- (ii) State what is meant by the term catalyst. 1

4. (continued)

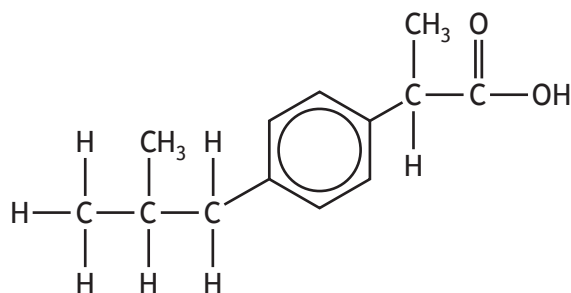
- (c) Many medicine molecules contain a chiral carbon which can affect their ability to work as a medicine.

A chiral carbon is any carbon atom that is attached to four different atoms or groups of atoms.

The chiral carbon in the L-DOPA molecule is circled.



Ibuprofen also has a chiral carbon in its molecule.



Circle the chiral carbon in the ibuprofen structure above.

1

(An additional structure, if required, can be found on *page 32*.)

- (d) An antagonist is a medicine that works by blocking the body's normal function. Painkillers are common examples of antagonists which work by blocking the body's response to pain.

Name the medicine, from the table on *page 12*, which would be described as an antagonist.

1

[Turn over



5. Petrol is a fuel used in some car engines.

(a) State what is meant by the term fuel.

1

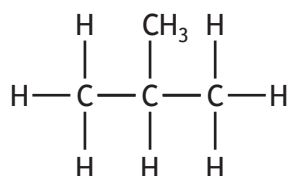
(b) The hydrocarbon iso-octane is added to petrol to improve performance in car engines.

(i) The systematic name for iso-octane is 2,2,4-trimethylpentane.

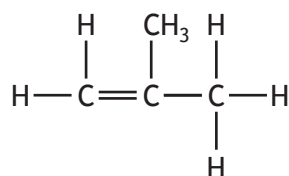
Draw a structure for iso-octane.

1

(ii) Iso-octane is produced by reacting together iso-butane and iso-butene.



iso-butane



iso-butene

State the test, including the result, that could be used to distinguish between a sample of iso-butane and iso-butene.

1



* X 8 1 3 7 5 0 1 1 4 *

5. (continued)

- (c) The flash point is the lowest temperature at which a fuel can catch fire.
The table shows the flash point of some alkanes.

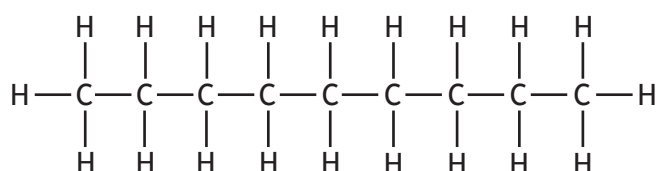
Alkane	Flash point (°C)
Hexane	−22
2,2,4-trimethylpentane	−12
Heptane	−4
2,2-dimethylhexane	−3
2-methylheptane	4
Octane	13

- (i) State the effect that increasing the number of branches in alkanes, containing the same number of carbon atoms, has on the flash point of the alkane.

1

- (ii) Predict the flash point, in °C, of the alkane shown.

1



[Turn over

5. (continued)

(d) Alcohols are commonly added to petrol to help them burn. These compounds are often described as oxygenates.

(i) Suggest why oxygenates help petrol burn.

1

(ii) The alcohols most commonly added to petrol are methanol and ethanol. The boiling points of methanol and ethanol are shown.

Alcohol	Boiling point (°C)
Methanol	65
Ethanol	78

Explain why ethanol has a higher boiling point than methanol.

2



* X 8 1 3 7 5 0 1 1 6 *

6. Fertilisers are substances which restore the elements essential for healthy plant growth to the soil.

Using your knowledge of chemistry, describe the chemistry of fertilisers.

3

[Turn over



* X 8 1 3 7 5 0 1 1 7 *

7. Radioactive decay occurs when unstable radioisotopes emit radiation to become more stable.

(a) State the part of the atom involved in radioactive decay.

1

(b) Phosphorus-32 is a radioisotope that is fed to mosquitoes to track how far they travel.



mosquito

A solution of sodium hydrogen phosphate, containing phosphorus-32, was fed to mosquitoes.

(i) Name the other elements present in sodium hydrogen phosphate.

1

(ii) State how the half-life of phosphorus-32, present in sodium hydrogen phosphate, compares to the half-life of a pure phosphorus-32 sample.

1

(c) Iodine-131 is a radioisotope which emits beta radiation and is used in medicine.

(i) The equation for the beta decay of iodine-131 is shown.



Name the element represented by X in the equation.

1



* X 8 1 3 7 5 0 1 1 8 *

7. (c) (continued)

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- (ii) A sample of iodine-131 had a mass of 40 g. After being left for 32 days the sample had a mass of 2.5 g.

Calculate the half-life, in days, of iodine-131.

2

- (d) Another radioisotope used in medicine is cobalt-60. Cobalt-60 has a half-life of 5 years.

A graph of the percentage of a cobalt-60 sample remaining over time is shown.

Complete the graph by adding:

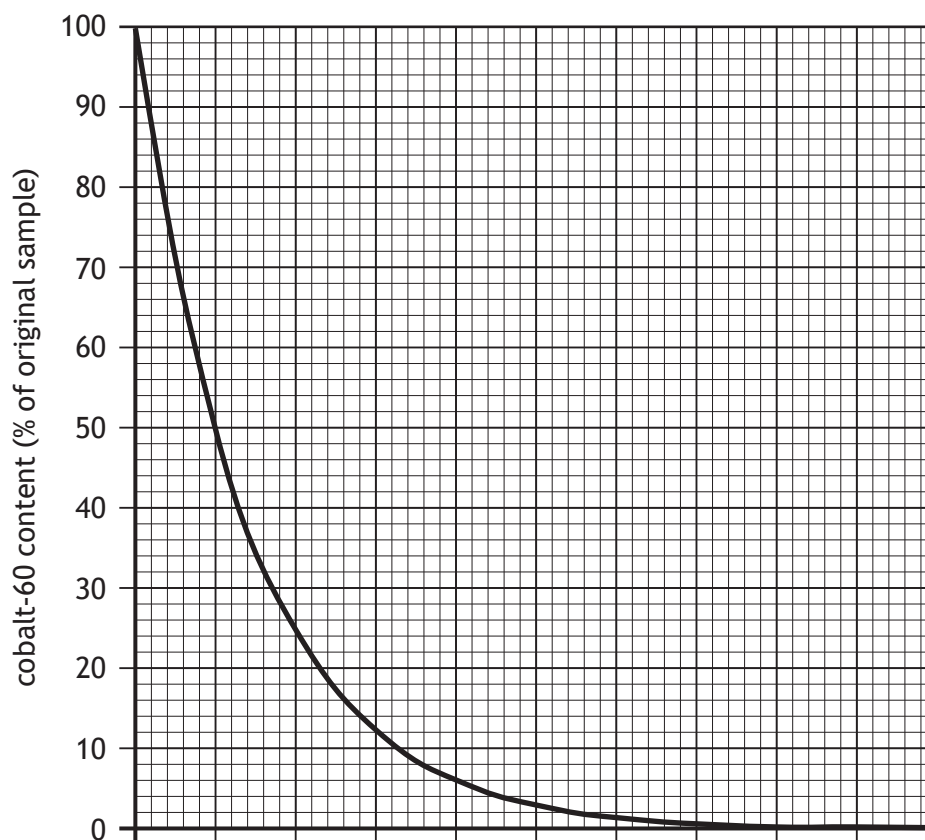
- (i) the scale to the x -axis

1

- (ii) the label to the x -axis.

1

(An additional graph, if required, can be found on *page 33*.)



8. Read the passage and answer the questions that follow.

The Mole

Atoms and molecules are so small that even one gram of a substance contains many trillions of atoms.

To simplify this, chemists measure the quantity of a chemical using a unit called the mole. This term was introduced in 1865 by the chemist August Wilhelm von Hofmann to describe the mass of a molecule, because the Latin term for mass is moles.

Another scientist, Amedeo Avogadro, took this one step further by defining one mole as the number of carbon atoms in 12 g of carbon, where each atom contains six protons and six neutrons. This number, known as Avogadro's number, is 6.02×10^{23} .

For example, one mole of water contains 6.02×10^{23} water molecules.

Avogadro's number is represented by the letter L. This can be used to calculate the number of atoms in one mole of molecules as shown in the table.

Molecular formula	Total number of atoms in molecule	Number of atoms in one mole
HCl	2	2L
NH ₃	4	4L

- (a) State the meaning of the Latin term 'moles'.

1

- (b) Write the nuclide notation for the carbon atom used to define Avogadro's number.

1



8. (continued)

(c) Calculate the mass, in grams, of 6.02×10^{23} water molecules, H_2O .

1

(d) Suggest why the number of electrons was not considered when calculating the mass of a carbon atom.

1

(e) Complete the table for methane, CH_4 .

1

Molecular formula	Total number of atoms in molecule	Number of atoms in one mole
CH_4		

[Turn over



* X 8 1 3 7 5 0 1 2 1 *

9. Propanol is an alcohol containing three carbon atoms.

(a) Propanol has two possible structures.

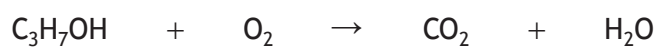
(i) State the term used to describe these two structures of propanol.

1

(ii) Draw **and** name one of the structures of propanol, C₃H₇OH.

2

(b) The equation for the combustion of propanol is shown.



Balance this equation.

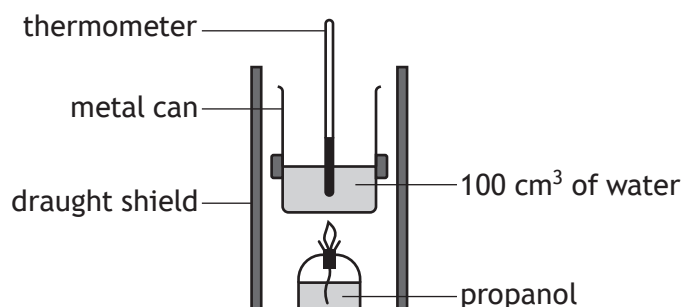
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9. (continued)

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- (c) A student carried out an investigation to calculate the energy released when burning propanol using the apparatus shown.



- (i) Explain why the use of a draught shield improves the accuracy of the student's results.
- (ii) The student burned propanol three times and recorded the temperature change of the water each time.

Experiment	1	2	3
Temperature change of the water (°C)	23	20	20

- (A) Calculate the average temperature change of the water, in °C, for the student's three experiments.
- (B) Using your answer to part (c) (ii) (A), calculate the average energy, in kJ, absorbed by the water in the student's investigation.



10. Sodium carbonate, Na_2CO_3 , is a white, water-soluble compound.

(a) One of the many uses of sodium carbonate is in dishwasher tablets.

Dishwasher tablets must be soluble in water.

(i) State the term used to describe liquids, like water, in which substances dissolve to form solutions.

1

(ii) Dishwasher tablets are often coated in a water-soluble plastic known as poly(ethenol) which is formed by addition polymerisation.

(A) Name the monomer used to make poly(ethenol).

1

(B) Suggest why ethanol could **not** be used as a monomer in an addition polymerisation reaction.

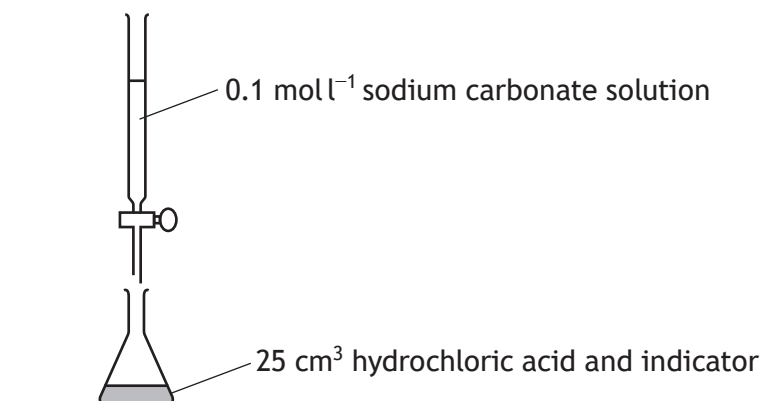
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10. (continued)

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MARGIN

- (b) An experiment to determine the concentration of a hydrochloric acid solution using a solution of sodium carbonate is shown.

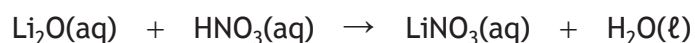


- (i) State the name given to this technique. 1
- (ii) Suggest why an indicator is added to the flask. 1
- (iii) The average volume of sodium carbonate, Na₂CO₃, used in the experiment was 19.0 cm³.
Sodium carbonate, Na₂CO₃, reacts with hydrochloric acid, HCl, in the ratio shown.
- $$1 \text{ mol Na}_2\text{CO}_3 : 2 \text{ mol HCl}$$
- Calculate the concentration, in mol l⁻¹, of the hydrochloric acid, HCl, used in the experiment. 3
- Show your working clearly.

[Turn over



11. A neutralisation reaction between lithium oxide and nitric acid is shown.



(a) Name the salt produced in the reaction of lithium oxide and nitric acid.

1

(b) Calculate the mass, in grams, of lithium oxide, Li_2O , required to prepare 250 cm^3 of 0.5 mol l^{-1} solution.

3

Show your working clearly.

(c) Another neutralisation reaction is shown.



Name substance X.

1

[Turn over



* X 8 1 3 7 5 0 1 2 7 *

11. (continued)

(d) Neutralisation reactions release heat energy.

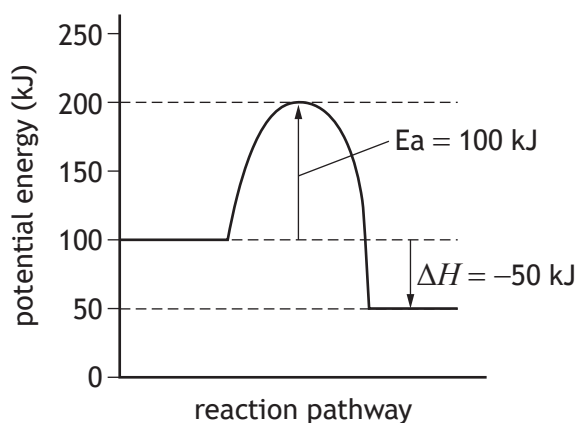
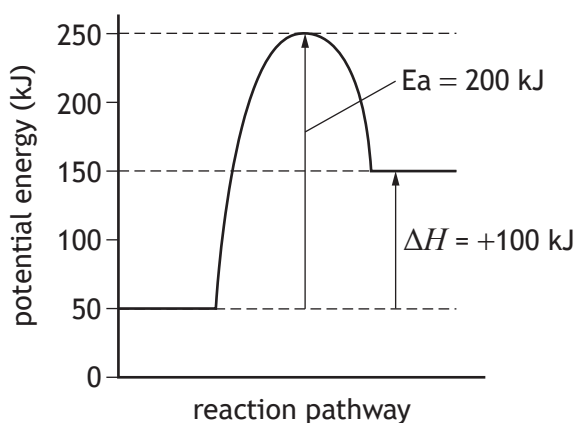
(i) State the term used to describe any reaction that releases heat energy.

1

(ii) Activation energy, E_a , and enthalpy change, ΔH , are energy values that are associated with chemical reactions.

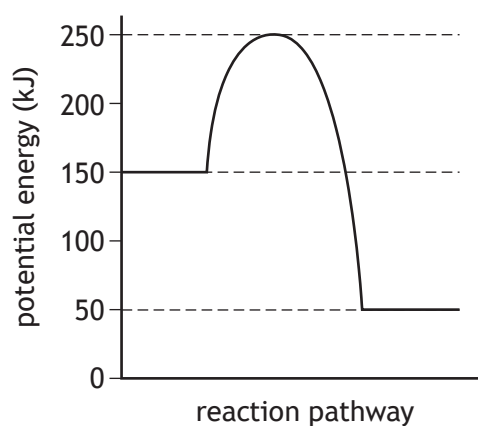
A potential energy diagram can be used to show these values.

(A) Two potential energy diagrams are shown below.



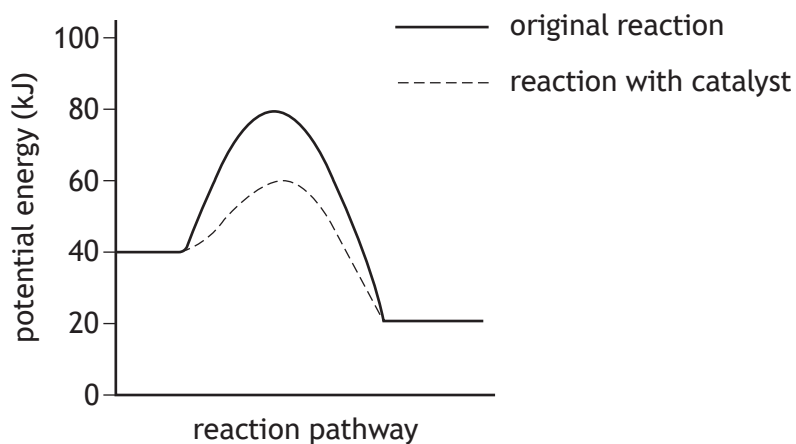
Calculate the enthalpy change, ΔH , in kJ, for the potential energy diagram shown below.

1



11. (d) (ii) (continued)

(B) When a catalyst is added to a reaction, the effect can also be seen on potential energy diagrams.



Circle the correct words to complete the table to show the effect of a catalyst on the activation energy, E_a , and the enthalpy change, ΔH .

2

Effect of catalyst on activation energy, E_a	<i>increase/decrease/no effect</i>
Effect of catalyst on enthalpy change, ΔH	<i>increase/decrease/no effect</i>

[Turn over

12. The reactivity series lists metals in order of reactivity.

Using your knowledge of chemistry, suggest experiment(s) that a student could carry out to place calcium, copper, tin and zinc into a reactivity series.

3

[END OF QUESTION PAPER]



* X 8 1 3 7 5 0 1 3 0 *



National
Qualifications
2025

2025 Chemistry

National 5

Question Paper Finalised Marking Instructions

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Marking instructions for each question

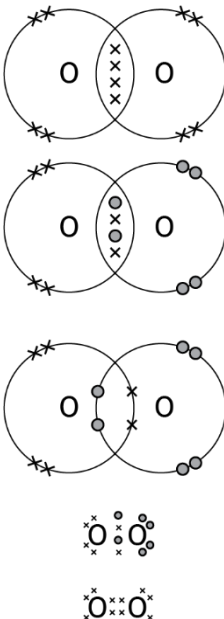
Section 1

Question	Answer	Mark
1.	A	1
2.	C	1
3.	D	1
4.	B	1
5.	B	1
6.	A	1
7.	D	1
8.	B	1
9.	A	1
10.	C	1
11.	D	1
12.	D	1
13.	C	1
14.	A	1
15.	B	1
16.	C	1
17.	B	1
18.	A	1
19.	D	1
20.	D	1
21.	A	1
22.	C	1
23.	C	1
24.	D	1
25.	B	1

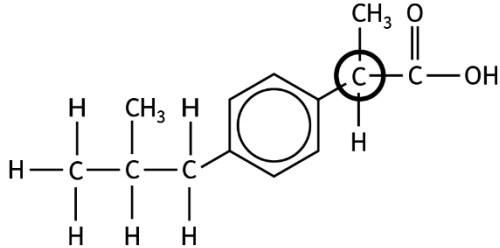
Section 2

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Burns with a pop.	1	Any mention of glowing splint would negate.
		(ii)	No stopper on conical flask. OR Bisecting line on end of delivery tube indicates a blockage. OR Graduation marks should be on measuring cylinder. OR Measuring cylinder should be labelled.	1	Any one correct response for 1 mark. If more than one is given, then all must be correct. Annotation on the diagram is acceptable if correct.
		(iii)	Evaporation	1	Accept a correct description e.g. heat it. Do not accept burning in place of heat. Filtration should be awarded zero marks but does not negate a correct answer.
	(b)	(i)	1.72 - 1.76 (cm³ s⁻¹) (2 marks) Partial marking 1 mark awarded for concept of change in volume/change in time. $\frac{86}{50}$ $\frac{87}{50}$ $\frac{88}{50}$ (1 mark)	2	The unit is not required but, if given, must be correct. This marking instruction must only be applied a maximum of once per paper. Accept cm³/s or unit in words. Do not accept cm³/s⁻¹ or sec. The mark for a final answer can only be awarded if the concept of change in volume/change in time is correct ie incorrect values from the graph used (volumes chosen must correspond to chosen times). An answer out with the range of 1.72 - 1.76 must be supported by working.

Question			Expected response	Max mark	Additional guidance
1.	(b)	(ii)	Line must start at zero and be less steep than the line given in the question. If the line levels off then it must do so on or after 80 s with a final volume of 96 cm³.	1	Apply a ½ box tolerance to where the line starts and where the line levels off.
		(iii)	48 (cm ³)	1	The unit is not required but, if given, must be correct. This marking instruction must only be applied a maximum of once per paper.

Question			Expected response	Max mark	Additional guidance
2.	(a)		Must show all outer electrons 	1	Accept cross or dot or e or e- to represent electrons or a mixture of these. Accept petal diagram. Bonding electrons MUST be on the line or in the overlapping area. Element symbols do not need to be shown if the circles for the outer shell are shown. If inner electrons are shown they must be correct.
	(b)		Any element name or correct symbol of: Sulfur/Selenium/Tellurium/ Polonium/Livermorium.	1	
	(c)		As the temperature increases the solubility decreases. OR As the temperature decreases the solubility increases.	1	Zero marks awarded for an incorrect cause and effect. eg As the solubility increases the temperature decreases. OR As the solubility decreases the temperature increases.
	(d)	(i)	Angular	1	
		(ii)	Positive, electrons	1	

Question			Expected response	Max mark	Additional guidance
3.	(a)	(i)	$ \begin{array}{cc} \text{H} & \text{Cl} \\ & \\ \text{C} & = & \text{C} \\ & \\ \text{H} & \text{H} \end{array} $	1	
		(ii)	Diatomic	1	Any mention of compound/element/atoms would negate a correct answer.
	(b)	(i)	Bars not points (1 mark) All bars correct (within a ½ box tolerance) (1 mark)	2	
		(ii)	The PVC would melt. OR The melting point of PVC is too low/less than 100 °C/less than the boiling point of water.	1	Zero marks awarded for 'it has a low/lower melting point' on its own, but this does not negate a correct answer.

Question			Expected response	Max mark	Additional guidance
4.	(a)		Carboxyl	1	Zero marks awarded for -COOH OR Carboxylic.
	(b)	(i)	Hydrogenation	1	
		(ii)	A substance that speeds up a chemical reaction (but can be recovered chemically unchanged at the end of the reaction).	1	Use of element / compound in place of substance is not accepted.
	(c)			1	
	(d)		Ibuprofen	1	

Question			Expected response	Max mark	Additional guidance
5.	(a)		Substance that burns/combusts to release energy/heat.	1	
	(b)	(i)	$ \begin{array}{ccccccc} & \text{H} & & \text{CH}_3 & & \text{H} & & \text{CH}_3 & & \text{H} \\ & & & & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - \text{H} \\ & & & & & & & & & \\ & \text{H} & & \text{CH}_3 & & \text{H} & & \text{H} & & \text{H} \end{array} $	1	Accept mixture of shortened and full structural formula. Zero marks awarded for C ₈ H ₁₈
		(ii)	Isobutene will decolourise bromine solution. OR Isobutane will not decolourise bromine solution.	1	Accept bromine/bromine water/ bromine solution but do not accept bromide or Br. Zero marks awarded for 'goes clear' however if given in addition to a correct answer it does not negate. If starting colour is given it must be correct eg orange/yellow/red/orange-brown/red-brown or brown. Award zero marks if the candidate implies it is the hydrocarbon that is being decolourised.
	(c)	(i)	(increasing the number of branches) Lowers (the flash point).	1	Zero marks awarded for an incorrect cause and effect. eg As the flash points lowers the number of branches increases. Zero marks awarded for decreasing the number of branches increases the flash point.
		(ii)	29 - 31 (°C)	1	The unit is not required but, if given, must be correct. This marking instruction must only be applied a maximum of once per paper.

Question			Expected response	Max mark	Additional guidance
5.	(d)	(i)	They contain/provide/supply oxygen. OR They ensure complete combustion .	1	Candidates response must indicate that the oxygenates contain oxygen.
		(ii)	Ethanol/it has bigger/stronger/more forces (of attraction). (1 mark) Between molecules or mention of intermolecular attractions. (1 mark) ----- If neither of these two points are given, a maximum of 1 mark can be awarded for - ethanol/it is bigger/has more (carbon or hydrogen) atoms/longer carbon chain.	2	The term bond is only acceptable if it is specifically identified as between the molecules or used with the term intermolecular. Mention of breaking bonds/bonds within molecule or chain/breaking carbon to carbon or carbon to hydrogen bonds or more bonds cannot gain the second mark but does not negate the first mark. Candidates can be awarded the full/partial marks if they correctly explain why methanol has a lower boiling point, but methanol must be stated in their answer. More bonds in the compound is not sufficient to imply a larger molecule but does not negate.

Question			Expected response	Max mark	Additional guidance
6.			This is an open ended question. 1 mark: The student has demonstrated a limited understanding of the chemistry involved. The candidate has made some statement(s) that is/are relevant to the situation, showing that at least a little of the chemistry within the context is understood. 2 marks: The student has demonstrated a reasonable understanding of the chemistry involved. The student makes some statement(s) that is/are relevant to the situation, showing that the context is understood. 3 marks: The maximum available mark would be awarded to a student who has demonstrated a good understanding of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the context. This does not mean the answer has to be what might be termed an “excellent” answer or a “complete” one.	3	

Question			Expected response	Max mark	Additional guidance
7.	(a)		The nucleus	1	
	(b)	(i)	Sodium, hydrogen and oxygen. OR Na, H and O	1	Phosphorus is not required but would not negate a correct answer. Phosphate would negate a correct answer. Accept H ₂ /O ₂ .
		(ii)	Stays the same	1	
	(c)	(i)	Xenon/Xenon-131/Xe/Xe-131 OR $^{131}\text{Xe} / ^{131}_{54}\text{Xe} / _{54}\text{Xe}$	1	
		(ii)	8 (days) (2 marks) Partial Marking 4 half lives (1 mark) If incorrect number of half lives is calculated then 1 mark can be awarded for 32 divided by the candidates number of half lives provided the arithmetic is correct.	2	The unit is not required but, if given, must be correct. This marking instruction must only be applied a maximum of once per paper.
	(d)	(i)	Scale - going up in increments of 5 starting at zero.	1	A minimum of two numbers must be shown on the scale. The zero does not need to be shown.
		(ii)	Time and Years (both required).	1	

Question			Expected response	Max mark	Additional guidance						
8.	(a)		Mass	1	Zero marks awarded for ‘mass of a molecule’ but this does not negate a correct answer.						
	(b)		$^{12}_6\text{C}$	1	Do not penalise if the atomic number and mass number is written on the right hand side of the symbol.						
	(c)		18 (g)	1	The unit is not required but, if given, must be correct. This marking instruction must only be applied a maximum of once per paper.						
	(d)		Electrons (they) have no/negligible mass.	1							
	(e)		<table><tr><th>Molecular formula</th><th>Total number of atoms in molecule</th><th>Number of atoms in one mole</th></tr><tr><td>CH₄</td><td>5</td><td>5L</td></tr></table>	Molecular formula	Total number of atoms in molecule	Number of atoms in one mole	CH ₄	5	5L	1	Accept 3.01×10 ²⁴ in place of 5L The L in the table must be uppercase.
Molecular formula	Total number of atoms in molecule	Number of atoms in one mole									
CH ₄	5	5L									

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	Isomers	1	
		(ii)	$\begin{array}{ccccc} & \text{H} & & \text{H} & & \text{H} \\ & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{OH} \\ & & & & & \\ & \text{H} & & \text{H} & & \text{H} \end{array}$ AND propan-1-ol (2 marks) OR $\begin{array}{ccccc} & \text{H} & & \text{OH} & & \text{H} \\ & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{H} \\ & & & & & \\ & \text{H} & & \text{H} & & \text{H} \end{array}$ AND propan-2-ol (2 marks) ----- Partial Marking 1 mark can be awarded for either a correct name or correct structure.	2	Accept mixture of shortened and full structural formula. Accept - 1-propanol in place of propan-1-ol - 2-propanol in place of propan-2-ol Ignore - the omission of one H atom, from a carbon, in full structural formula provided the bond is shown. OR - one carbon to hydrogen bond missing provided the hydrogen is shown. Connectivity to the hydroxyl group must be to the oxygen.
	(b)		$\text{C}_3\text{H}_7\text{OH} + 4.5 \text{O}_2 \rightarrow 3 \text{CO}_2 + 4 \text{H}_2\text{O}$	1	Accept correct multiples. If equation is rewritten, all formulae must be correct and correct format. Ignore state symbols if given.
	(c)	(i)	To prevent heat loss (to the surroundings).	1	
		(ii) (A)	21 (°C)	1	The unit is not required but, if given, must be correct. This marking instruction must only be applied a maximum of once per paper.

Question			Expected response	Max mark	Additional guidance
9.	(c)	(ii) (B)	8.78/8.778/8.8/9 (kJ) (3 marks) ----- Partial Marking Use of correct concept of $E_h = cm\Delta T$ with $c = 4.18$ (1 mark) To be awarded this concept mark, candidates do not specifically need to write $cm\Delta T$. The concept mark is awarded for multiplying three values, one of which must be 4.18 0.1 and answer from (c)(ii)(A). (1 mark) A further mark can be awarded for the candidate's calculated answer only if the mark for the concept has been awarded. (1 mark)	3	Allow follow through from (c)(ii)(A). No units required but a maximum of two marks can be awarded if an incorrect unit is given. This marking instruction must only be applied a maximum of once per paper. 8778 J can be awarded 3 marks if used with 4180 and the correct unit is given.

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	Solvent	1	
		(ii) (A)	Ethenol	1	
		(ii) (B)	It does not have a carbon-to-carbon double bond/it only contains single carbon-to-carbon bonds. OR It is saturated/it is not unsaturated.	1	
	(b)	(i)	Titration	1	
		(ii)	To indicate the end point/to show when the reaction is neutral/complete.	1	

Question			Expected response	Max mark	Additional guidance
10.	(b)	(iii)	0.152 / 0.15 / 0.2 (mol l⁻¹) (3 marks) ----- Method A 1 mark for the correct number of moles of sodium carbonate. ie n = CV = 0.1 × 0.019 = 0.0019 mol (1 mark) 1 mark for calculating the moles of hydrochloric acid by correctly applying the mole ratio. ie 2 × candidate's calculated number of moles of Na₂CO₃ (1 mark) 0.0038 mol of HCl on its own. (2 marks) 1 mark for calculating the concentration of the hydrochloric acid. ie C = n/V using candidate's calculated number of moles of HCl and 0.025 (1 mark) ----- Method B $\frac{0.1 \times 19}{1} = \frac{C_2 \times 25}{2}$ (1 mark) (1 mark) C₂ × 12.5 = 1.9 or C₂ × 25 = 3.8 (this step on its own gets 2 marks) OR $\frac{0.1 \times 0.019}{1} = \frac{C_2 \times 0.025}{2}$ (1 mark) (1 mark) C₂ × 0.0125 = 0.0019 OR C₂ × 0.025 = 0.0038 (this step on its own gets 2 marks)	3	No units required but a maximum of two marks can be awarded if an incorrect unit is given. This marking instruction must only be applied a maximum of once per paper. Accept mol l⁻¹ or mol/l but not mol/l⁻¹ or mol⁻¹ or mol l For method A Candidates should not be penalised if 19 and 25 (volumes of sodium carbonate and hydrochloric acid) are both expressed in cm³. If candidate only calculates number of moles of sodium carbonate the volume must be in litres to be awarded 1 mark. If candidate correctly multiplies their calculated number of moles of sodium carbonate by 2 the mark for the mole ratio can be awarded. If concentration of incorrect chemical is calculated then max = 1 mark. For method B using relationship shown in the data book. 1 mark is awarded for the correct pairings of volume (in the same unit) and concentration. 1 mark is awarded for the correct mole ratio being applied. 1 mark is awarded for the correct arithmetic. This mark can only be awarded if either of the first two marks have been awarded.

Question			Expected response	Max mark	Additional guidance
11.	(a)		Lithium nitrate	1	
	(b)		3.75/3.8 (g) (3 marks) ----- Partial Marks can be awarded for a maximum of two of the following three steps: 1 mark for correctly calculating the number of moles of lithium oxide. ie $n = CV = 0.25 \times 0.5 = 0.125 \text{ mol}$ (1 mark) 1 mark for correctly calculating the GFM of lithium oxide. ie GFM = 30 (1 mark) 1 mark for calculating the mass of lithium oxide. ie $m = n \times \text{GFM}$ using candidates calculated moles of lithium oxide and candidates calculated GFM (1 mark)	3	No units required but a maximum of two marks can be awarded if an incorrect unit is given. This marking instruction must only be applied a maximum of once per paper. An incorrect number of moles or GFM of lithium oxide must be supported by working.
	(c)		Copper carbonate OR CuCO_3	1	
	(d)	(i)	exothermic	1	
	(d)	(ii) (A)	–100 (kJ)	1	Negative value must be shown. The unit is not required but, if given, must be correct. This marking instruction must only be applied a maximum of once per paper.
		(ii) (B)	Decrease (1 mark) No Effect (1 mark)	2	

Question			Expected response	Max mark	Additional guidance
12.			This is an open ended question. 1 mark: The student has demonstrated a limited understanding of the chemistry involved. The candidate has made some statement(s) that is/are relevant to the situation, showing that at least a little of the chemistry within the context is understood. 2 marks: The student has demonstrated a reasonable understanding of the chemistry involved. The student makes some statement(s) that is/are relevant to the situation, showing that the context is understood. 3 marks: The maximum available mark would be awarded to a student who has demonstrated a good understanding of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the context. This does not mean the answer has to be what might be termed an “excellent” answer or a “complete” one.	3	

[END OF MARKING INSTRUCTIONS]

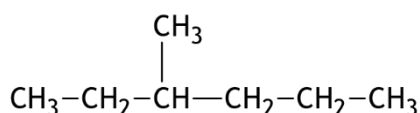
General marking principles for National 5 Chemistry

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

- (a) Marks for each candidate response must always be assigned in line with these general marking principles and the detailed marking instructions for this assessment.
- (b) If a specific candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.
- (c) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

A guiding principle in marking is to give credit for correct chemistry rather than to look for reasons not to award marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon.

Although the punctuation is not correct, '3, methyl-hexane' should gain the mark.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule.

The results are shown in the table.

<i>Structural formula</i>	<i>pH</i>
CH ₃ COOH	1.65
CH ₂ ClCOOH	1.27
CHCl ₂ COOH	0.90
CCl ₃ COOH	0.51

State how the strength of the acids is related to the number of chlorine atoms in the molecule.

Although not completely correct, an answer such as 'the more Cl₂, the stronger the acid' should gain the mark.

- (d) There are no half marks awarded.
- (e) Candidates must respond to the 'command' word as appropriate and may be required to write extended answers in order to communicate fully their knowledge and understanding.

- (f) Marks should be awarded for answers that have incorrect spelling or loose language **as long as the meaning of the word(s) is conveyed**. **Example:** Answers like ‘distilling’ (for ‘distillation’) and ‘it gets hotter’ (for ‘the temperature rises’) should be accepted.

However, the example below would not be given any credit, as an incorrect chemical term, which the candidate should know, has been given.

Example: If the correct answer is ‘ethene’, and the candidate’s answer is ‘ethane’, this should not be accepted.

- (g) A correct answer followed by a wrong answer should be treated as a cancelling error and no marks should be awarded.

Example: State what colour is seen when blue Fehling’s solution is warmed with an aldehyde.

The answer ‘red, green’ gains no marks.

If a correct answer is followed by additional information which does not conflict, the additional information should be ignored, whether correct or not.

Example: State why the tube cannot be made of copper.

If the correct answer is related to a low melting point, ‘Copper has a low melting point and is coloured grey’ would not be treated as having a cancelling error.

- (h) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units if required) on its own.

The partial marks shown in the marking scheme are for use when working is given but the final answer is incorrect. An exception is when candidates are asked to ‘Find, by calculation’, when full marks cannot be awarded for the correct answer without working.

- (i) In most questions units are not required. However, if the candidate writes units then they must be correct. An incorrect unit would not be acceptable and one mark would not be awarded.

This marking instruction must only be applied a maximum of once per paper.

- (j) Where the marking instructions specifically allocate a mark for units in a calculation, this mark should not be awarded if the units are incorrect or missing. Missing or incorrect units at intermediate stages in a calculation should be ignored.

- (k) As a general rule, where a wrong numerical answer (already penalised) is carried forward to another step, credit will be given provided the result is used correctly. The exception to this rule is where the marking instructions for a numerical question assign separate ‘concept marks’ and an ‘arithmetic mark’. In such situations, the marking instructions will give clear guidance on the assignment of partial marks.

- (l) Ignore the omission of one H atom from a full structural formula provided the bond is shown or one carbon to hydrogen bond missing provided the hydrogen is shown.

- (m) A symbol or correct formula should be accepted in place of a name **unless stated otherwise in the marking instructions**.

- (n) When formulae of ionic compounds are given as answers it will only be necessary to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, no marks should be awarded.

- (o) If an answer comes directly from the text of the question, no marks should be awarded.

Example: A student found that 0.05 mol of propane, C_3H_8 burned to give 82.4 kJ of energy.

Name the type of enthalpy change which the student measured.

No marks should be awarded for 'burning' since the word 'burned' appears in the text.

- (p) Unless the question is clearly about a non-chemistry issue, eg costs in industrial chemical process, a non-chemical answer gains no marks.

Example: Suggest why the (catalytic) converter has a honeycomb structure.

A response such as 'to make it work' may be correct but it is not a chemical answer and the mark should not be awarded.